Algebra For Physics Students

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Preface

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Chapter 1. Set theory

This chapter includes necessary knowledge about the set theory. I will not talk anything about the ZFC axioms, or any axiom system of constructing sets. I assume that we already know what are sets and some basic operations, such as the union, the intersection, the complement, etc.

1.1 Sets

The first notion I want to mention is the **ordered pair**. For two elements a, b, the ordered pair (a, b) is defined as,

$$(a, b) := \{\{a, b\}, \{b\}\}. \tag{1.1.1}$$

We can see that $(a, b) \neq (b, a)$ if $a \neq b$, that's why it's called an ordered pair. We can easily generalize it to an ordered collection of n elements recursively as

$$(a_1, a_2, \ldots, a_n) := \{\{a_1, a_2, \ldots, a_n\}\} \cup (a_2, a_2, \ldots, a_n), \tag{1.1.2}$$

which is called an n-**tuple**. By this definition, we can see if we change any two nonequivalent elements, we will get another n-tuple. Thus, this definition gives us an ordered object from the pure set theory. Now we define the **Cartesian product** of two sets X, Y as

$$X \times Y := \{(x, y) | x \in X, y \in Y\}. \tag{1.1.3}$$

And for n sets, we can also define

$$X_1 \times X_2 \times \cdots \times X_n = \{(x_1, x_2, \ldots, x_n) | x_1 \in X_1, x_2 \in X_2, \ldots, x_n \in X_n\}. \tag{1.1.4}$$

In particular, if the n sets are all the same, we write it as,

$$X^n \equiv X \times X \times \cdots \times X.$$

1.2 Relations

A **relation** in the set X is a subset of $X \times X$. For example, we can define the $<$ relation in the set $X = \{1, 2, 3\}$ as the subset

$$<_X = \{(1, 2), (1, 3), (2, 3)\}. \tag{1.2.1}$$

We usually denote $(x, y) \in \triangleright \subset X \times X$, for $\triangleright$ a relation in X, as $x \triangleright y$.

There is a special kind of relations called **equivalence relation**. We call a relation $\sim$ an equivalence relation in the set X if:

(1) $x \sim x, \forall x \in X$;

(2) $x \sim y \Rightarrow y \sim x$, $\forall x, y \in X$;

(3) $x \sim y$ and $y \sim z \Rightarrow x \sim z$, $\forall x, y, z \in X$.

If we define an equivalence relation in the set X, we call the subset

$$[x] := \{y \in X | y \sim x\}. \tag{1.2.2}$$

The equivalence class of x. We can prove if $x, y \in X$, $[x] \cap [y] = \emptyset$ or $[x] = [y]$. Then we can define the quotient set $X/\sim$ as the set of all equivalence classes of X.

1.3 Maps

A **map** from the set X to Y is a subset of $f \subset X \times Y$, such that

$$\forall x \in X, \exists! p \in f, \pi_X(p) = x. \tag{1.3.1}$$

Usually we denote the map f from X to Y, as $f : X \to Y$, and $(x, y) \in f$ as $f : x \mapsto y$ or $f(x) = y$. A map $f : X \to Y$ is said to be injective if:

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2, \tag{1.3.2}$$

to be surjective if:

$$\forall y \in Y, \exists x \in X, \text{such that } f(x) = y. \tag{1.3.3}$$

A map is called bijective if it is both injective and surjective.

Chapter 2. Vectors

2.1 Vector spaces

$\mathbb{F} = \mathbb{R}$ of $\mathbb{C}$.

2.2 Inner products

$\langle \cdot, \cdot \rangle : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$

2.3 Linear maps

2.4 Tensors

A multilinear map is a map f from the Cartesian product of vector spaces $\mathcal{V}_1, \mathcal{V}_2, \ldots, \mathcal{V}_n$ to $\mathcal{W}$, all vector spaces are over the same field $\mathbb{F}$, satisfying,

$$\begin{aligned} &f(\mathbf{v}_1, \mathbf{v}_2, \cdots, a\mathbf{v}_i + b\mathbf{v}_i', \cdots, \mathbf{v}_n) \\ &= af(\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_i, \cdots, \mathbf{v}_n) + bf(\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_i', \cdots, \mathbf{v}_n) \end{aligned} \tag{2.4.1}$$

We have already known what are a vector space $\mathcal{V}$ and its dual space $\mathcal{V}^*$. The tensor of type (r, s) of vector space $\mathcal{V}$ as a multilinear mapping

$$T^r_s : \underbrace{\mathcal{V}^* \times \cdots \times \mathcal{V}^*}_{r \text{ times}} \times \underbrace{\mathcal{V} \times \cdots \times \mathcal{V}}_{s \text{ times}} \to \mathbb{F}. \tag{2.4.2}$$

The set containing all (r, s)-tensor is denoted as $\mathscr{T}^r_s$. The components of a (r, s)-tensor T^r_s are defined as,

$$T^{\mu_1 \cdots \mu_r}{}_{\nu_1 \ldots \nu_s} := T^r_s(\epsilon^{\mu_1}, \ldots, \epsilon^{\mu_r}, \mathbf{e}_{\nu_1}, \ldots, \mathbf{e}_{\nu_s}). \tag{2.4.3}$$

We can show

$$T^r_s = T^{\mu_1 \cdots \mu_r}{}_{\nu_1 \ldots \nu_s} \mathbf{e}_{\mu_1} \otimes \cdots \otimes \mathbf{e}_{\mu_r} \otimes \epsilon^{\nu_1} \otimes \cdots \otimes \epsilon^{\nu_s}. \tag{2.4.4}$$

Thus, sometimes we also consider the (r, s)-tensors as elements in the vector space

$$\underbrace{\mathcal{V} \otimes \cdots \otimes \mathcal{V}}_{r \text{ times}} \otimes \underbrace{\mathcal{V}^* \otimes \cdots \otimes \mathcal{V}^*}_{s \text{ times}}. \tag{2.4.5}$$

2.5 Complexification

For a vector space $\mathcal{V}$ over $\mathbb{R}$, the complexification of it is $\mathcal{V} \otimes \mathbb{C}$, here $\mathbb{C}$ is 2-dimensional real vector space, The basis for this space is,

$$\mathbf{e}_j \otimes 1, \quad \mathbf{e}_j \otimes i \tag{2.5.1}$$

where $\{e_j\}_{j=1}^{\dim \mathcal{V}}$ is the set of basis of $\mathcal{V}$. We define,

$$a(\mathbf{v} \otimes b) = \mathbf{v} \otimes (ab), \quad a, b \in \mathbb{C}. \tag{2.5.2}$$

Now the basis of $\mathcal{V}^{\mathbb{C}} = \mathcal{V} \otimes \mathbb{C}$ becomes

$$\mathbf{e}_j \otimes 1 \tag{2.5.3}$$

since $\mathbf{e}_j \otimes 1$ and $\mathbf{e}_j \otimes i$ are linear dependent now and the scalar becomes $\mathbb{C}$. For a complex vector space $\mathcal{V}$, we can define an anti-linear involution $\sigma : \mathcal{V} \to \mathcal{V}$, satisfying

$$\sigma^2 = Id \tag{2.5.4}$$

which is the meaning of an involution. An anti-linear map f can also be fully determined by the behavior of the basis. If we know $\mathbf{e}_j \mapsto f(\mathbf{e}_j)$, $j = 1, \ldots, n$, for any $\mathcal{V} \ni \mathbf{v} = a_i \mathbf{e}_1 + \cdots a_n \mathbf{e}_n$, we have

$$f(\mathbf{v}) = a_1^* \mathbf{e}_1 + \cdots + a_n^* \mathbf{e}_n. \tag{2.5.5}$$

The simplest example of antilinear involution is the antilinear "identity" $Id(\mathbf{e}_j) = \mathbf{e}_j$. A non-trivial example in two-dimensional complex vector space is given by $\sigma(\mathbf{e}_1) = -\mathbf{e}_2, \sigma(\mathbf{e}_2) = -\mathbf{e}_1$. The subspace

$$\mathrm{Fix}(\sigma) := \{\mathcal{V} | \sigma(\mathbf{v}) = \mathbf{v}, \mathbf{v} \in \mathcal{V}\} \tag{2.5.6}$$

is

Chapter 3. Operators

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Chapter 4. Algebras

An algebra A is a vector space over $\mathbb{F}$ with an additional binary operation, $\star : A \times A \to A$, satisfying

$$\begin{aligned}(x+y)\star z =& a\star z + y\star z\\ z\star(x+y) =& z\star a + z\star y\\ (ax)\star(by) =& (ab)(x\star y)\end{aligned} \tag{4.0.1}$$

for all $a, b \in \mathbb{F}$ and $x, y, z \in A$. We may add more axioms to get associative, unital, commutative algebra.

4.1 Exterior algebra

In section 2.4, we have defined tensors.

$$\omega = \frac{1}{p!}\sum_{i_1,\cdots,i_p}^{D} \omega_{i_1\cdots i_p} e^{i_1}\wedge\cdots\wedge e^{i_p} \tag{4.1.1}$$

For

$$\Omega = \omega\wedge\eta \tag{4.1.2}$$

$$\Omega_{i_1\cdots i_{p+q}} = \frac{(p+q)!}{p!q!}\omega_{i_1\cdots i_p}\eta_{i_{p+1}\cdots i_{p+q}} = \binom{p+q}{p}\omega_{i_1\cdots i_p}\eta_{i_{p+1}\cdots i_{p+q}} \tag{4.1.3}$$

The parameter $\binom{p+q}{p}$ can be understood as the Shuffle sum

$$\Omega_{i_1\cdots i_{p+q}} = \sum_{\sigma\in Sh(p+q)} (-1)^{|\sigma|}\omega_{i_{\sigma_1}\cdots i_{\sigma_p}}\eta_{i_{\sigma_{p+1}}\cdots i_{\sigma_{p+q}}} \tag{4.1.4}$$

4.2 Clifford algebra

4.3 Lie algebra

4.3.1 Complexification of the Lie algebra

For a complex Lie algebra $\mathfrak{g}$ with basis $\{T_k\}_{k=1}^d$ and structure constants f^{ijk} we can consider it as a real Lie algebra $\mathfrak{g}_{\mathbb{R}}$ with basis $\{T_k^{re}, T_k^{im}\}_{k=1}^d$, where T_k^{re}, T_k^{im} are not T_k, iT_k in general. But since they are in the same Lie algebra, we need the closure of Lie bracket as,

$$[T_i^{re}, T_j^{re}] = f^{ijk} T_j^{re}, \quad [T_i^{re}, T_j^{im}] = f^{ijk} T_k^{im}, \quad [T_i^{im}, T_j^{im}] = -f^{ijk} T_k^{re}. \tag{4.3.1}$$

The right-hand sides of these three equation are all well-defined.

4.3.2 Real forms of $\mathfrak{sl}(2;\mathbb{C}) \oplus \mathfrak{sl}(2;\mathbb{C})$

The Cartan-Weyl basis of $\mathfrak{sl}(2,\mathbb{C})$ are

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \tag{4.3.2}$$

And for two $\mathfrak{sl}(2,\mathbb{C})$, we may use $\{H_L, E_L, F_L\}$ and $\{H_R, E_R, F_R\}$ respectively. Another often used basis are,

$$S_1 = \frac{1}{2}\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}, \quad S_2 = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad S_3 = \frac{1}{2}\begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \tag{4.3.3}$$

which are just $S_k = \frac{\sigma_k}{2i}$.

$$\begin{aligned} S_1^{(L)} &= -\frac{i}{2}(E_L + F_L), \quad S_2^{(L)} = \frac{1}{2}(F_L - E_L), \quad S_3^{(L)} = -\frac{i}{2}H_L \\ S_1^{(R)} &= -\frac{i}{2}(E_R + F_R), \quad S_2^{(R)} = \frac{1}{2}(F_R - E_R), \quad S_3^{(R)} = -\frac{i}{2}H_R. \end{aligned} \tag{4.3.4}$$

If

$$\sigma(X, Y) = (-X^\dagger, -Y^\dagger), \tag{4.3.5}$$

For $\sigma(X, Y) = (X, Y)$, we have two such equations

$$\begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix} = \begin{pmatrix} -\alpha^* & -\gamma^* \\ -\beta^* & \alpha^* \end{pmatrix} \tag{4.3.6}$$

The general solution is,

$$\frac{a}{2}\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} + \frac{b}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + \frac{c}{2}\begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \equiv aJ_1 + bJ_2 + cJ_3 \tag{4.3.7}$$

for $a, b, c \in \mathbb{R}$. We can see,

$$[J_i, J_j] = \epsilon_{ijk} J_k. \tag{4.3.8}$$

Thus, we have the direct sum of two independent $\mathfrak{su}(2)$,

$$\mathrm{Fix}(\sigma) = \mathfrak{su}(2) \oplus \mathfrak{su}(2). \tag{4.3.9}$$

I write down the basis relation.

$$J_k^{(L)} = S_k^{(L)}, \quad J_k^{(R)} = S_k^{(R)} \tag{4.3.10}$$

If

$$\sigma(X, Y) = (\bar{X}, \bar{Y}) \tag{4.3.11}$$

$$\begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix} = \begin{pmatrix} \alpha^* & \beta^* \\ \gamma^* & -\alpha^* \end{pmatrix} \tag{4.3.12}$$

The solution is just

$$a\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + b\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \tag{4.3.13}$$

which represents the 2-dimensional traceless matrix, the generators of the group $SL(2, \mathbb{R})$. Thus, we have the direct sum of two independent $\mathfrak{sl}(2, \mathbb{R})$,

$$\mathrm{Fix}(\sigma) = \mathfrak{sl}(2; \mathbb{R}) \oplus \mathfrak{sl}(2; \mathbb{R}) \tag{4.3.14}$$

If

$$\sigma(X, Y) = (\bar{Y}, \bar{X}) \tag{4.3.15}$$

we have

$$\begin{pmatrix} \alpha_1 & \beta_1 \\ \gamma_1 & -\alpha_1 \end{pmatrix} \oplus \begin{pmatrix} \alpha_2 & \beta_2 \\ \gamma_2 & -\alpha_2 \end{pmatrix} = \begin{pmatrix} \alpha_2^* & \beta_2^* \\ \gamma_2^* & -\alpha_2^* \end{pmatrix} \oplus \begin{pmatrix} \alpha_1^* & \beta_1^* \\ \gamma_1^* & -\alpha_1^* \end{pmatrix}, \tag{4.3.16}$$

the general solution is

$$\begin{gathered} a_0(H_L + H_R) - a_1(iH_L - iH_R) + b_0(E_L + E_R) - b_1(iE_L - iE_R) + \\ c_0(F_L + F_R) - c_1(iF_L - iF_R) \in \mathrm{Fix}(\sigma). \end{gathered} \tag{4.3.17}$$

We define

$$H^{re}, E^{re}, F^{re} := H_L + H_R, E_L + E_R, F_L + F_R \tag{4.3.18}$$

$$H^{im}, E^{im}, F^{im} := iH_R - iH_L, iE_R - iE_L, iF_R - iF_L. \tag{4.3.19}$$

We can show this new set of basis satisfying condition (4.3.1) of $\mathfrak{sl}(2, \mathbb{C})$, thus

$$\mathrm{Fix}(\sigma) = \mathfrak{sl}(2; \mathbb{C})_{\mathbb{R}} \tag{4.3.20}$$

which is a 6-dimensional real Lie algebra. Another basis of $\mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}}$ is given by

$$\begin{aligned} J_1 &= -\frac{i}{2}(E^{re} + F^{re}), \quad J_2 = \frac{1}{2}(F^{re} - E^{re}), \quad J_3 = -\frac{i}{2}H^{re} \\ K_1 &= -\frac{i}{2}(E^{im} + F^{im}), \quad K_2 = \frac{1}{2}(F^{im} - E^{im}), \quad K_3 = -\frac{i}{2}H^{im}. \end{aligned} \tag{4.3.21}$$

Expressed in the conventions here, J_i is S_i^{re} and K_i is S_i^{im}. The commutation relation is given by,

$$\begin{aligned} [J_a, J_b] &= i\epsilon_{abc}J_c, \\ [J_a, K_b] &= i\epsilon_{abc}K_c, \\ [K_a, K_b] &= -\, i\epsilon_{abc}J_c, \end{aligned} \tag{4.3.22}$$

which is just the commutation relation of $\mathfrak{so}(1, 3)$. So $\mathfrak{so}(1, 3) \cong \mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}}$.

Chapter 5. Groups

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Chapter 6. Rings

6.1 Basics

A ring is a triple $(R, +, \cdot)$, where R is a set, $+$ and $\cdot$ are two binary operations on R, such that

(1) $(R, +)$ is an Abelian group, with identity element denoted by 0;

(2) $(R, \cdot)$ is a semigroup, with identity element denoted by 1;

(3) the distributive law, i.e. for all $x, y, z \in R$, $x(y+z) = xy + xz$ and $(y+z)x = yx + zx$.

6.2 Polynomial rings

Let R be a ring, the polynomial ring $R[x]$ is the set of all polynomials in x with coefficients in R. For example, if $R = \mathbb{Z}_5$, then an element in $\mathbb{Z}_5[x]$ is of the form $f(x) = [3] + [4]x + [2]x^2$. The addition and multiplication are defined as usual. We can also define the multivariate polynomial ring $R[x_1, x_2, \ldots, x_n]$.

We can assign a degree to each variable, which may be element of $\mathbb{Z}, \mathbb{Z}_2, \mathbb{Z}^n$ and many others, for example, $\deg x = 1, \deg y = 2$, then we define the degree of a monomial as the sum of the degree of each variable, for example, $\deg(x^2y^3) = 2\deg x + 3\deg y = 2 + 6 = 8$. The degree of a polynomial as the maximum degree of its monomials. For example, $\deg(4x + 2x^3) = 3$.

If every monomial in a polynomial has the same degree, we call it a homogeneous polynomial. For example, when $\deg x = \deg y = 1$, $f(x, y) = 4x^2 + 2xy + y^2$ is homogeneous of degree 2.

6.3 Symmetric polynomials

The symmetric polynomials are polynomials invariant under the permutation of variables, which form a subring of $\mathbb{Z}[x_1, \ldots, x_n]$. We denote it as

$$\Lambda_n = \mathbb{Z}[x_1, \ldots, x_n]^{S_n}.$$

Λ_n is graded by the degree of polynomials, and the homogeneous symmetric polynomials of degree k is denoted as Λ_n^k. For $\alpha \in \mathbb{Z}^n$, we denote the monomial $x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ by x^α. And we will use x to represent the tuple $(x_1, \ldots, x_n)$. Let λ be a partition with $\ell(\lambda) \leqslant n$. The polynomial m_λ is defined as

$$m_\lambda(x) = \sum_\alpha x^\alpha, \tag{6.3.1}$$

where α runs through all distinct permutations of λ, or more precisely, all distinct permutations of

$$(\lambda_1, \lambda_2, \ldots, \lambda_{\ell(\lambda)}, \underbrace{0, \ldots, 0}_{n-\ell(\lambda)}).$$

For example, if $\lambda = (2, 1)$ and $n = 3$, then

$$m_\lambda(x_1, x_2, x_3) = x_1^2 x_2 + x_1^2 x_3 + x_2^2 x_1 + x_2^2 x_3 + x_3^2 x_1 + x_3^2 x_2.$$

The rth elementary symmetric polynomials are defined as

$$e_r(x) = \sum_{1 \leqslant i_1 < \cdots < i_r \leqslant n} x_{i_1} \cdots x_{i_r} = m_{(1^r)}. \tag{6.3.2}$$

The rth complete symmetric polynomials are defined as

$$h_r(x) = \sum_{1 \leqslant i_1 \leqslant \cdots \leqslant i_r \leqslant n} x_{i_1} \cdots x_{i_r} = \sum_{|\lambda| = r} m_\lambda. \tag{6.3.3}$$

The rth power sum are defined as

$$p_r(x) = \sum_i x_i^r = m_{(r)}. \tag{6.3.4}$$